

Mini MES (Manufacturing Execution System)

Operational Manual, Role Reference & Process Flow Guide

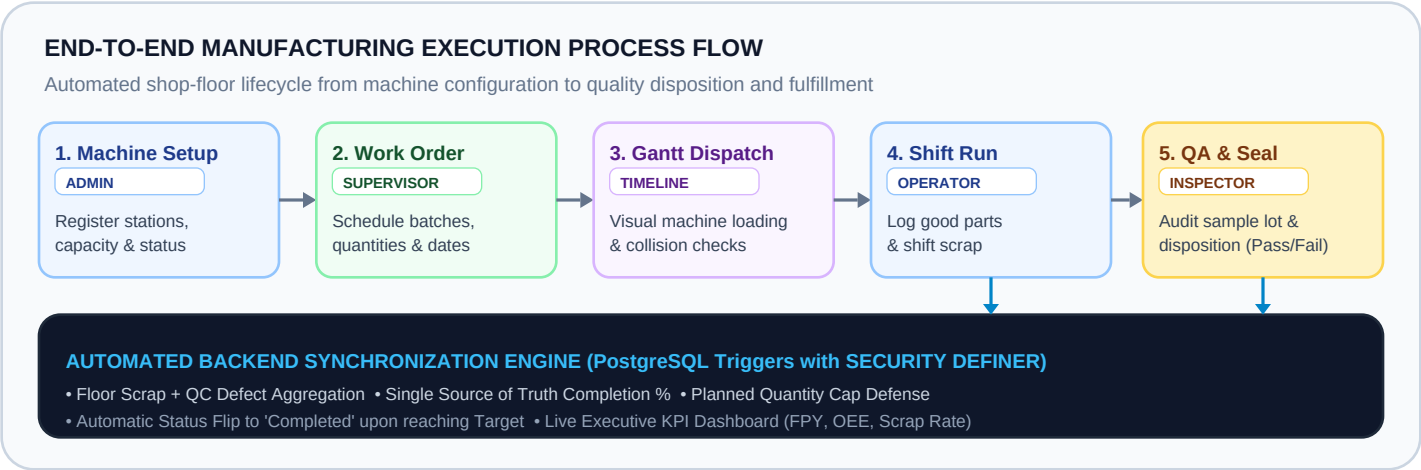
1. Executive Overview & System Purpose

Mini MES is a real-time **Manufacturing Execution System** engineered for discrete and batch production environments. It acts as the single source of operational truth on the factory floor, orchestrating work orders, machine stations, operator shifts, and quality audits.

- **Work Order Life-Cycle:** Planning, machine assignment, progress tracking, and automated completion.
- **Machine Scheduling & Gantt Timeline:** Visual horizontal timeline plotted across workstations and dates, complete with overload conflict warning indicators and drag-to-reschedule.
- **Shift Output & Floor Scrap:** Fast operator logging of good units versus rejected scrap with planned quantity defense guards.
- **Quality Assurance (QA) Audits:** Lot sample inspections, defect classifications, and dispositioning.
- **Database-Enforced Security:** PostgreSQL Row-Level Security (RLS) and elevated SECURITY DEFINER recalculation triggers.

2. Manufacturing Execution Process Flowchart

The visual diagram below maps how orders, machines, and shift logs flow seamlessly from creation to final executive analytics:



3. User Roles & Clean Login Credentials

All credentials use clean alphanumeric passwords with no special characters. Powered by Supabase Auth with PostgreSQL RLS:

Role	Login Email	Password	System Permissions & Scope
Admin	admin@factory.com	Admin123	Unrestricted control. Create/edit/delete orders, register machines, Gantt schedule timeline, production logs, QA audits, diagnostics.
Supervisor	supervisor@factory.com	Supervisor123	Production scheduling. Plan and release work orders, Gantt timeline dispatch, drag-reschedule batches, manage shift logs, view analytics.
Operator	operator@factory.com	Operator123	Floor execution. Submit shift production logs (good units vs scrap), monitor active jobs and station statuses. (Read-only on schedule/orders).
Quality Inspector	quality@factory.com	Quality123	Quality auditing authority. Log inspection sample sizes, defect classifications, and record lot disposition (Pass / Fail / Conditional).

4. End-to-End Operational Walkthrough Example

Scenario: Manufacturing a batch of **500 Electric Motor Shafts 25mm** (Work Order: WO-2026-8178) on workstation CNC-01.

Step 1: Create & Schedule the Work Order (Supervisor)

1. Log in with supervisor@factory.com / Supervisor123.
2. Navigate to **Work Orders** → click **+ New Work Order**.
3. Enter batch parameters:
 - **Order Number:** WO-2026-8178
 - **Product Name:** Electric Motor Shaft 25mm
 - **Machine:** CNC-01 | **Target Quantity:** 500
 - **Dates:** Start = Today, Due = Next Friday.
4. Click **Create Work Order**. The batch initializes at 0% in **Pending** status.

Step 2: Inspect Gantt Timeline & Workstation Loading (Supervisor)

1. In the sidebar, click **Gantt Schedule**.
2. The horizontal span bar for WO-2026-8178 appears across CNC-01's row.
3. **Conflict Detection:** If another order is booked on CNC-01 during overlapping dates, an amber Conflict (2) warning badge automatically flags the collision and stacks the bars in clear vertical lanes.
4. **Interactive Reschedule:** Drag the bar horizontally to adjust start/due dates in real-time, or drag the right edge handle to extend due date.

Step 3: Record Shift Production Output (Operator)

1. Sign out and log in as operator@factory.com / Operator123.
2. Click **Production Logs** → click **+ Log Production**.
3. Select Work Order WO-2026-8178 and enter:
 - **Produced Qty (Passed Units):** 200
 - **Rejected / Scrap Qty:** 5
 - **Shift:** Shift A (Morning)
4. Click **Save Entry**.

Automated System Reaction: The PostgreSQL SECURITY DEFINER trigger updates the work order record to 200 Produced, 5 Rejected (40%) and advances status to **In Progress**. On the Gantt chart, the bar displays a mini progress-fill spanning 40%.

Step 4: Conduct Quality Assurance Sample Audit (Quality Inspector)

1. Sign out and log in as quality@factory.com / Quality123.
2. Go to **Quality** → click **+ New Inspection**.
3. Link to WO-2026-8178:
 - **Sample Inspected:** 25 units | **Defect Count:** 1 unit
 - **Classification:** Dimensional Tolerance | **Result:** Pass
4. Total rejected count synchronizes to **6 units** (5 floor scrap + 1 QC defect).

Step 5: Final Shift Completion & Automatic Seal (Operator)

1. Sign in as operator@factory.com.
 2. Log remaining batch output: Produced: 294 (or 300), Scrap: 0.
- Automated Sealing:** Once good units reach the planned 500 target, the trigger immediately flips status to **Completed (100%)**.

Step 6: Real-Time Intelligence Review (Management)

Open **Live Dashboard**. Management monitors:

- **Fulfillment Rate:** 100% (Good Units Only).
- **Scrap Rate:** 1.2% | **First Pass Yield (FPY):** 98.8% conforming output.
- **Top Orders Graph:** WO-2026-8178 reflects complete delivery.

5. Technical Architecture & Deployment

Database Trigger Privileges: The PostgreSQL trigger function `public.recalculate_work_order_totals()` is defined with **SECURITY DEFINER SET search_path = public, pg_temp**. This bypasses Row Level Security when updating work_orders totals, guaranteeing that Operator shift logs update production counts and status seamlessly.

Vercel Production Deployment:

1. Push latest codebase to GitHub: `git push origin main`.
2. In Vercel, import project with preset **Vite** (build command: `npm run build`).
3. Configure Environment Variables: `VITE_SUPABASE_URL` and `VITE_SUPABASE_ANON_KEY`.
4. Client-side Single Page Application (SPA) routing is handled automatically by `vercel.json`.